

西南财经大学本科期末考试试题册（A）

（2023—2024学年第1学期）

以下各项由命题教师填写：

课程名称：数据结构（英） 命题教师：陈中普 王海林

适用对象（年级专业）：2022级信息管理与信息系统、2022级物流管理、
2022级金融数学

使用试题的任课教师姓名：陈中普 王海林

试题说明：

- 1、考试类型：闭卷[☒] 开卷[]
- 2、本套试题共 5 道大题，共 5 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 字典[] 等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证于桌面左上角，以备查验。
 2. 拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
 4. 所有答案均需填写在答题纸上，答在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

- We assume that the height of an empty tree is -1.
- The Node class in a singly linked list has attribute *data* and *next*, in which *data* is the element it is holding, and *next* is the link to the next node.
- The Node class in a binary tree has attribute *key*, *left*, and *right*, in which *key* is the data it is holding, and *left* and *right* are links to its left and right child, respectively.

Part One: Multiple Choices (2 marks each; 20 marks in total)

(Note: Each question has only one correct choice.)

1. Suppose an initially empty Stack S has executed a total of 30 push operations, 12 top operations, and 10 pop operations. Note that during those pop operations, 5 times of Empty error were caught. What is the current size of S?
 - A. 8
 - B. 15
 - C. 20
 - D. 25
2. What is time complexity of the following code?


```
s = 0
for i in range(N, 2):
    for j in range(10):
        s += i * j
```

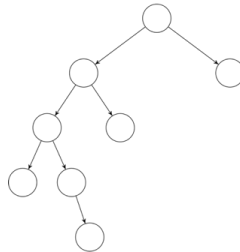
 - A. $O(N)$
 - B. $O(N^2)$
 - C. $O(1)$
 - D. $O(\log N)$
3. Given a BST, node p appears before node q during the in-order traversal. Which one is NOT possible in terms of their relationships?
 - A. q is p's parent
 - B. q is p's right child
 - C. q is p's right sibling
 - D. q is p's parent's parent
4. Given a binary tree with 255 nodes, what is the minimal height of this tree?
 - A. 6
 - B. 7
 - C. 8
 - D. 9

5. Which factors do affect the average search performance while hashing?

i. Load factor; ii. Hash function; iii. Collision-handing scheme

- A. Only i and ii
- B. Only i and iii
- C. Only ii and iii
- D. i, ii and iii

6. Please assign keys 1, 3, 4, 6, 8, 11, 20, 100 to the binary search tree below. What is the left child of the root?



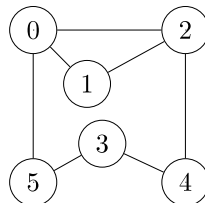
- A. 6
- B. 8
- C. 11
- D. 20

7. Consider a min-heap stored in an array [5, 8, 12, 19, 28, 20, 15, 22]. After inserting 3, what is the final underlying array of this heap?

- A. [3, 5, 12, 8, 28, 20, 15, 22, 19]
- B. [3, 5, 12, 19, 20, 15, 22, 8, 28]
- C. [3, 8, 12, 5, 20, 15, 22, 28, 19]
- D. [3, 12, 5, 8, 28, 20, 15, 22, 19]

8. Consider an undirected graph below, in which adjacent lists of each vertex are:

adj[0] = [2, 1, 5]; adj[1] = [0, 2]; adj[2] = [0, 1, 3, 4]; adj[3] = [5, 4, 2]; adj[4] = [3, 2]; adj[5] = [3, 0]



Which vertex is lastly visited in a depth-first search from vertex 0?

- A. 5
- B. 4
- C. 3
- D. 2

9. Bob would like to hash [1, 10, 11, 20, 22, 33, 21, 40, 24, 35] to an array with size of 10 using separate chaining with the hash function $h(x) = x \% 10$. What is the size of the longest chain?

- A. 2
- B. 3
- C. 4
- D. 5

10. Which statement is correct?

- A. Big O is always used to describe the worst-case performance.
- B. A stack is also called a list.
- C. A binary search tree is balanced.
- D. Linked list has more storage overhead compared to an array.

Part Two: True or False (2 mark each; 20 marks in total)

1. Given a singly linked list T only with a head node, the time complexity of inserting a new node at the end of it is $O(N)$, where N is size of T .
2. Deque follows the first-in first-out (FIFO) policy.
3. Given an undirected graph implemented with a adjacency matrix, the time complexity of checking whether vertex w is adjacent to v is $O(1)$.
4. Given a BST, the minimal key is always at the leftmost.
5. $O(\max\{f(n) + g(n)\}) = O(f(n) + g(n))$.
6. The root of a red-black tree is red.
7. If we would like to design an efficient queue based on a singly linked list with both head and tail nodes, then dequeue operation can be implemented by removing tail.
8. Alice hashes a sequence of $[2, 3, 5, 8, 28, 18]$ into an array A with linear probing using hash function $h(x) = x \% 10$. Key 18 is at position 0 if the size of A is 12.
9. Breadth-first search algorithms on a graph are based on stacks.
10. A circular linked list is a type of linked list where the last node points back to the first node.

Part Three: Short Q&A (6 marks each; 24 marks in total)

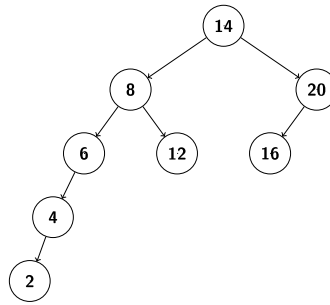
(Note that only key points are required and try to keep your answers short)

1. In order to implement a stack based an array, why is it inefficient to push and pop at the front of the array?
2. To design a circularly queue based on an array with initial capacity 4, in which we use head to denote the position of the first element in the queue. After enqueue(1), enqueue(2), enqueue(3), dequeue(), enqueue(4), what happens if we execute enqueue(5) and enqueue(6) sequentially?
3. Given an integer i , the following recursive algorithm is to compute the sum of each digit of i , where “/” is floor integer division. For example, $sum(123)$ is 6. However, there is a bug in code. How to fix this bug?

Algorithm 1: $sum(i)$

return $i \% 10 + sum(i / 10)$

4. How to make the AVL tree below re-balanced?



Part Four: Design and Analysis (36 marks in total)

(Note that pseudo-code and even plain English are allowed.)

- Given a singly linked list with only the head, it is called small if either more than half of its elements is less than 5 or it is empty. For example, 3->10->1->4 is small, while 3->10->8 is not small. Please design *is_small()* to check whether it is small. (10 marks) (Hint: it should return either True or False)

```

class LinkedList:
    def __init__(self):
        self._head = None
    def is_small(self):
        pass # implement this
  
```

- Given a binary search tree with a root node, please design an algorithm *contains()* to check whether it contains a target key. (10 marks) (Hint: it should return either True or False)

```

class BST:
    def __init__(self):
        self._root = None
    def contains(self, key):
        pass # implement this
  
```

- Given a singly sorted linked list only with the head node. Please design an algorithm *insert()* to add a new element while keeping it sorted. For example, after inserting 7, 1->4->8->20 becomes 1->4->7->8->20. (10 marks)

```

class SortedLinkedList:
    def __init__(self):
        self._head = None
    def insert(self, key):
        pass # implement this
  
```

- You can choose either A or B. (6 marks)

A. Consider a social network in which each vertex (an integer) denotes a user, and each edge denotes the friendship between two users. Please design an algorithm *friends()* to return the numbers of mutual direct and indirect friends of *u* and *v*.

```

class SocialNetwork:
  
```

```

def __init__(self, vertices):
    self.vertices = vertices
    self.adjacency_list = {vertex: [] for vertex in range(vertices)}
def add_edge(self, u, v):
    self.adjacency_list[u].append(v)
    self.adjacency_list[v].append(u)
def friends(self, u, v):
    pass # implement this

```

- B. We say a function $f(N)$ is polynomially bounded if $f(N) = O(N^k)$ for some constant k . $\lceil \cdot \rceil$ is the function to get a ceil integer; and operator $!$ is to compute factorial. Is the function $\lceil \log N \rceil!$ polynomially bounded?

(Hint: $n!$ approximates to $\sqrt{2\pi n}(\frac{n}{e})^n$)